"Signal-Peptide Tuning of mRNA-encoded α-Galactosidase A (GLA) to Maximize Secretion and Cross-Correction for Fabry Disease"

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Abstract

Fabry disease is caused by a deficiency of α-galactosidase A, encoded by the *GLA* gene, leading to progressive accumulation of glycosphingolipids in multiple tissues. mRNA based enzyme expression represents a promising therapeutic approach. However, efficient secretion and functional delivery of the encoded enzyme remain key challenges. The N terminal signal peptide plays a critical role in directing newly synthesized proteins into the secretory pathway, yet its influence on the performance of mRNA encoded α-galactosidase A has not been systematically investigated.

This study aims to evaluate how variation in signal peptide sequence affects the expression, secretion, and functional competence of mRNA encoded α-galactosidase A. A panel of *GLA* mRNA constructs incorporating different N terminal signal peptides will be designed and expressed in HEK293 cells. Intracellular expression and secretion kinetics will be quantified using enzymatic activity assays and protein analysis. Secreted enzyme from selected variants will be further characterized for post translational processing, glycosylation, and stability under physiologically relevant conditions.

By integrating molecular design with biochemical and functional validation, this work seeks to establish a rational framework for signal peptide optimization in mRNA based enzyme therapies. The outcomes are expected to inform the development of improved therapeutic strategies for Fabry disease and may be broadly applicable to other lysosomal storage disorders.

1. Introduction

Anderson–Fabry disease, commonly called Fabry disease (FD), is a rare X-linked disorder caused by mutations in the *GLA* gene. These mutations reduce or eliminate the activity of the enzyme α-galactosidase A (α-Gal A), leading to the buildup of globotriaosylceramide (Gb3) in lysosomes (Lenders & Brand, 2021). Over time, this accumulation causes progressive damage to multiple organs, including the kidneys, heart, and nervous system, and contributes to a wide range of symptoms such as neuropathic pain, cardiac complications, and strokes (Umer & Kalra, 2023).

Current treatment options include enzyme replacement therapy (ERT) and pharmacological chaperone therapy. ERT works by supplementing the body with functional α-Gal A through regular intravenous infusions. While it can reduce substrate accumulation, its effectiveness varies between patients, requires lifelong high-dose administration, and can sometimes trigger immune reactions. Chaperone therapy, on the other hand, stabilizes certain mutant forms of α-Gal A, but it only works for patients with specific responsive mutations.

These limitations highlight the need for new treatment approaches that can provide consistent enzyme activity, reduce treatment frequency, and be effective across a broader range of patients. mRNA-based therapeutics are emerging as a promising option. By delivering mRNA that encodes functional α-Gal A, these therapies enable cells to produce the enzyme directly, offering a potentially more efficient and flexible approach. This strategy may overcome some of the key challenges of existing therapies, such as enzyme instability, immunogenicity, and mutation-specific restrictions. Recent preclinical work demonstrates that systemic delivery of α-Gal A mRNA via lipid nanoparticles can restore enzyme activity, reduce Gb3 and related substrate accumulation in tissues, and sustain effects for weeks after a single dose in animal models supporting the rationale for exploring mRNA therapy for FD (Zhu et al., 2019a). With the rapid progress in mRNA technology and its proven success in other areas of medicine, exploring mRNA-based therapies could open a new opportunity for the treatment of Fabry disease.

2. Overview of Fabry Disease

Lysosomal storage diseases (LSDs) represent a group of more than 70 rare inherited metabolic disorders caused by defects in lysosomal function. These conditions arise mainly due to single-gene mutations that impair lysosomal enzyme activity, leading to the abnormal accumulation of substrates within affected cells (Sheth et al., 2023).

Among these disorders, Fabry disease (FD; OMIM 301500) is an X-linked genetic condition in which males are more severely affected due to their hemizygous genetic makeup, whereas females are typically carriers and may exhibit milder or variable disease manifestations (Thompson et al., 2025). FD is caused by mutation in *GLA* gene present in the short arm of chromosome X (Xq22.1), which cannot produce sufficient α-Gal A enzyme (Ioannou et al., 2001). This enzyme is a homodimeric glycoprotein that catalysis the cleavage of α-1,4-galactosidic bond, a critical step in the breakdown of a specific glycosphingolipid called globotriaosylceramide (Gb3 or GL3, also known as ceramide trihexoside) and its deacylated metabolite lyso-globotriaosylceramide (lyso-Gb3) inside lysosomes (Van Der Veen et al., 2020).

FD leads to several other disease complications that mainly affect heart, kidney, brain and skin (Kubo, 2017), among which, cardiac involvement is the main reason that causes disrupted quality of life and death in affected patients (Zarate & Hopkin, n.d.). Historically, this disease was first described by Willliam Anderson and Johannes Fabry, who independently reported in 1898 (Anderson, 1898; Fabry, 1898)

FD presents itself through two mechanisms: classical and nonclassical. Classical FD emerges symptomatically from childhood due to no or little residual α-Gal A. Non classical FD arises during adulthood owing to presence of partial α-Gal A activity, with the heart being the most affected organ (Germain & Linhart, 2024).

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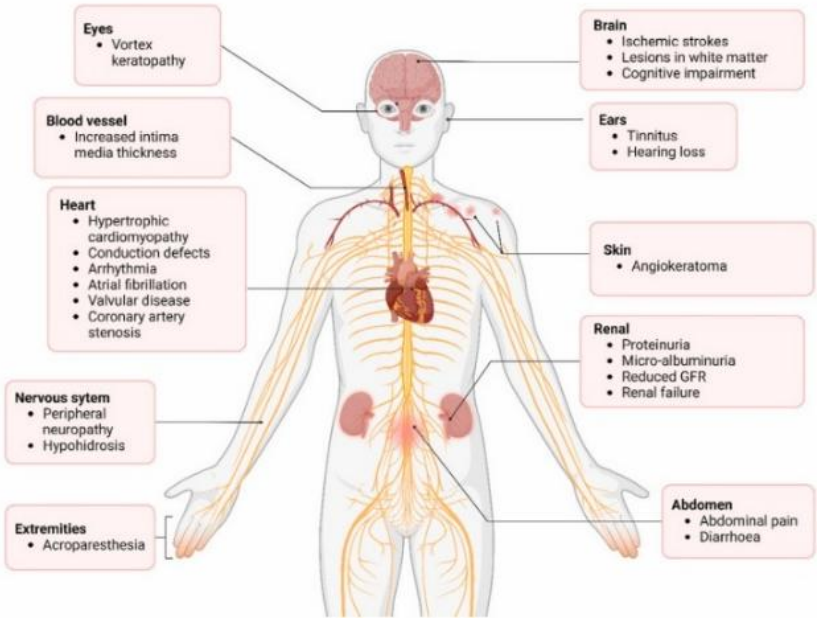


Figure 1. Clinical Symptoms of Fabry Disease (Raj et al., 2024)

3. Current Therapeutic Landscape

Different therapeutics approaches have been tried to overcome this rare genetic condition, two of which have been approved by the U.S. Food and Drug Administration (FDA) and the European Medicines Agency (EMA). These therapies include enzyme replacement therapy (ERT) and pharmacological chaperone therapy that can help reduce the buildup of fatty substances in the kidneys, heart, and blood vessels (Lenders et al., 2025).

3.1 Enzyme Replacement Therapies (ERT)

In ERT, patients with Fabry disease receive the functional form of α-Gal A. This therapeutic enzyme is produced by cloning the human *GLA* gene into an appropriate expression vector, followed by expression in a suitable host system, and subsequent purification and characterization. Two approved drugs namely, Agalsidase-α (Replagal, Shire, Lexington, MA) and Agalsidase-β (Fabrazyme, Sanofi-Genzyme, Cambridge, MA) are administered intravenously once every week at the recommended dosage of 0.2 mg/kg body weight (Gómez-Cerezo et al., 2025).

Agalsidase alfa and Agalsidase beta have an identical α-Gal A amino-acid sequence and comparable enzymatic activity. The structural difference lies in post-translational glycosylation, resulting from their production in different expression systems (Sakuraba et al., 2006). Agalsidase beta contains a higher proportion of phosphorylated oligomannose glycans and higher mannose-6-phosphate content, while Agalsidase alfa shows a relatively higher proportion of complex-type oligosaccharides with lower phosphorylation. Agalsidase beta also displays greater overall sialylation, leading to enhanced binding to the mannose-6-phosphate receptor and increased cellular uptake in vitro (Lee, 2003).

Pegunigalsidase-α, a PEGylated and cross-linked α-Gal A is an enzyme therapy produced in genetically modified *Nicotiana tabacum* plant cells. With a longer half-life and effective Gb3 clearance, Pegunigalsidase alfa exhibits non-inferior renal effectiveness to Agalsidase beta. however, hypersensitivity reactions and immune-complex-mediated nephritis in certain patients may restrict its usage (Germain & Linhart, 2024).

Moss-α-Gal, a moss-derived variant of human α-galactosidase with a uniform N-glycosylation pattern and more than 90% mannose-terminated glycans. A single intravenous dose of moss-α-galactosidase A was safe and well tolerated and produced a sustained reduction in urinary Gb3, indicating effective renal targeting, while plasma Gb3 and lyso-Gb3 showed only modest changes (Hennermann et al., 2019).

3.2 Chaperone therapy

The first oral pharmacological chaperone medication authorised for the treatment of Fabry disease is Migalastat (marketed under the brand name Galafold®). By binding to the α-Gal A enzyme's active site, migalastat enhances its stability and lessens endoplasmic reticulum degradation. In Fabry disease patients with amenable *GLA* mutations, migalastat therapy reduces left ventricular hypertrophy, stabilises renal function, and lowers or maintains low plasma lyso-Gb3 levels with good tolerability (Weidemann et al., 2022).

3.3 Substrate Reduction Therapy

Substrate reduction therapy focuses on inhibiting glucosylceramide synthesis, which leads to a decrease in the accumulation of GL3 in cells. The oral drug lucerastat, administered in addition to enzyme replacement therapy, was well tolerated over a 12-week period and resulted in a marked 55% reduction in plasma Gb3 levels, without adversely affecting renal or cardiac function (Guérard et al., 2018).

Another oral drug, venglustat, has also been developed as a potential treatment for Fabry disease. Venglustat produced sustained reductions in disease-related glycosphingolipids, including plasma GL-3 and lyso-GL-3, indicating effective substrate reduction during long-term treatment. Renal and cardiac parameters remained largely stable, and the drug was generally well tolerated, supporting its potential as an oral therapy for Fabry disease (Deegan et al., 2023).

3.4 Limitations of Existing Treatment Approaches

Current therapies for Fabry disease, including ERT and oral chaperone therapy, have significantly improved patient outcomes, yet still have few limitations. One major drawback of ERT is the development of anti-drug antibodies (ADAs). Approximately 40% of male patients develop neutralizing antibodies against infused α-Gal A, which reduce enzyme uptake and diminish therapeutic efficacy over time (Gómez-Cerezo et al., 2025). Secondly, ERT is limited by its uneven distribution across tissues. While liver cells can take up the enzyme effectively, key target cells such as heart muscle cells, kidney podocytes, and vascular cells often receive insufficient amounts to achieve meaningful therapeutic effect. In addition, the enzyme cannot cross the blood-brain barrier, which limits its effectiveness in addressing potential central nervous system involvement (Azevedo et al., 2020). A significant limitation of enzyme-based therapies is the rapid degradation and clearance of administered enzymes in the body leading

to frequent dosing schedules, typically every two weeks (De La Fuente et al., 2021). This can be challenging for FD patients requiring lifelong intravenous infusions.

Migalastat is effective only in patients with specific GLA variants that can respond to the treatment, limiting its applicability to about 35-50% of the Fabry population (Perretta et al., 2025a). Even among eligible patients, responses to chaperone therapy vary, with differences seen in enzyme activity, biomarker reduction, and organ-specific outcomes. Therefore, careful patient selection and long-term follow-up are necessary, and chaperone therapy cannot currently be regarded as a universal substitute for other Fabry disease treatment options (Nowicki et al., 2024).

Clinical studies of SRT agents have primarily demonstrated biomarker reductions rather than definitive improvement in hard clinical endpoints like renal failure, stroke, or cardiomyopathy, limiting conclusions on long-term disease modification. Furthermore, lucerastat clinical trials showed that primary clinical endpoints like neuropathic pain did not significantly improve despite successful substrate reduction, suggesting that biochemical correction may not be sufficient to properly treat illness symptoms (Guérard et al., 2018; Deegan et al., 2023).

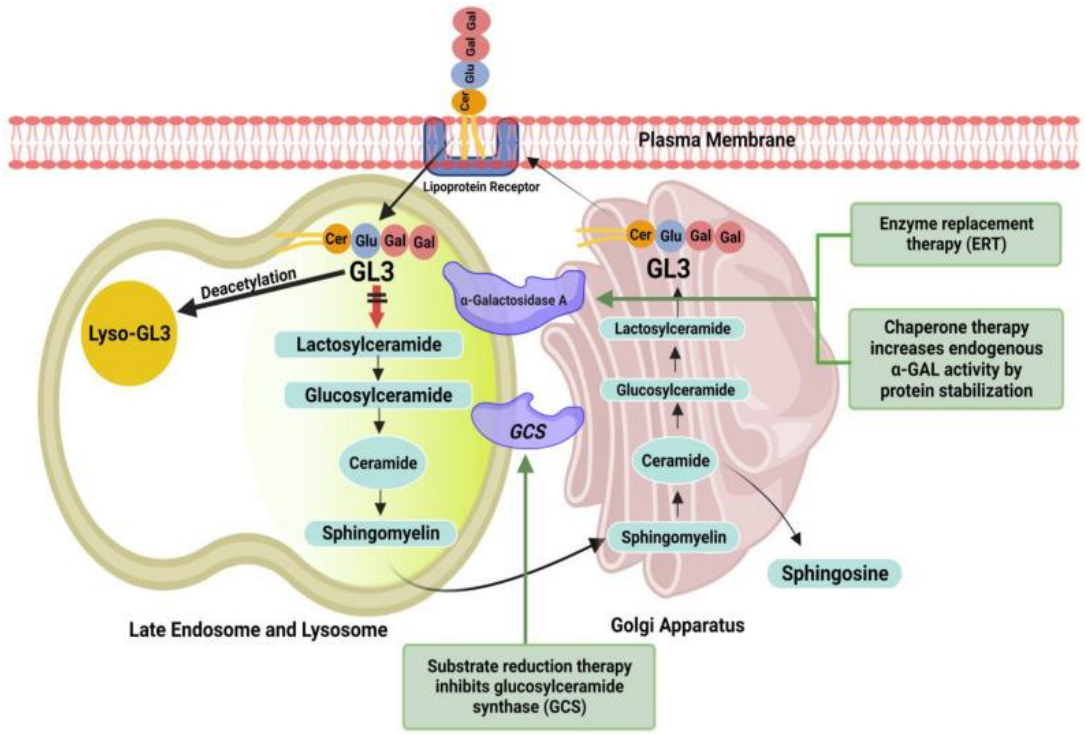


Figure 2 Overview of the ceramide pathway and therapeutic targets (Umer & Kalra, 2023)

Table 1. Comparison of Different therapies for Fabry Disease

Therapy	Pharmacological mechanism	Presently available therapeutics	Advantages	Disadvantages	Reference
Enzyme replacement therapy (ERT)	Intravenous recombinant α-galactosidase A degrades Gb3 and Lyso-Gb3 in tissues	Agalsidase (Fabrazyme), Agalsidase (Replagal), Pegunigalsidase alfa (Elfabrio)	Clinically established	Lifelong intravenous administration & neutralizing antibodies development	(Lenders et al., 2025)
Pharmacological chaperone therapy	Chaperone stabilizes specific mutant α-galactosidase A and enhances lysosomal activity	Migalastat (Galafold)	Orally administered; Good long-term safety and patient compliance.	Effective only for certain mutations	(Germain et al., 2016; Hughes et al., 2017)
Substrate reduction therapy (SRT)	Inhibition of glucosylceramide synthase reduces Gb3 synthesis	Lucerastat & Venglustat	Oral administration improves compliance and allows central nervous system exposure.	Clinical efficacy data are limited and large phase III trials are lacking.	(Perretta et al., 2025)

4. mRNA Therapeutics: Scientific Background

Messenger RNA (mRNA) based therapeutics represent a versatile platform for transient protein expression, enabling cells to synthesize functional proteins without altering the host genome. Therapeutic mRNA is designed to mimic endogenous eukaryotic mRNA and collectively regulate stability, translation efficiency, and intracellular localization (Van Lint et al., 2013).

4.1 mRNA structure and processing

Messenger RNA (mRNA) is a single-stranded, negatively charged RNA molecule that carries genetic information from DNA to ribosomes for protein synthesis. Its structure includes a 5′ cap, a 5′ untranslated region (5′ UTR), a coding sequence (open reading frame), a 3′ UTR, and a poly(A) tail (Faure et al., 2016). The 5′ cap and poly(A) tail protect mRNA from degradation and enhance translation by interacting with ribosomal and initiation factors. UTRs and internal

RNA structures regulate translation efficiency, stability, and localization, while the coding sequence determines the amino acid sequence of proteins (Kim et al., 2022)

Translation begins when initiation factors recognize the 5′ cap and recruit the 40S ribosomal subunit to form the pre-initiation complex, which scans the 5′ UTR to locate the start codon, after which the 60S subunit joins to begin elongation. The balance between translation and decay is tightly controlled, as mRNAs are degraded through deadenylation-dependent pathways, endonuclease-mediated cleavage, or decapping mechanisms to maintain appropriate protein levels (Shyu et al., 2008) . Together, these elements ensure precise control of gene expression and are key to designing effective mRNA-based therapeutics.

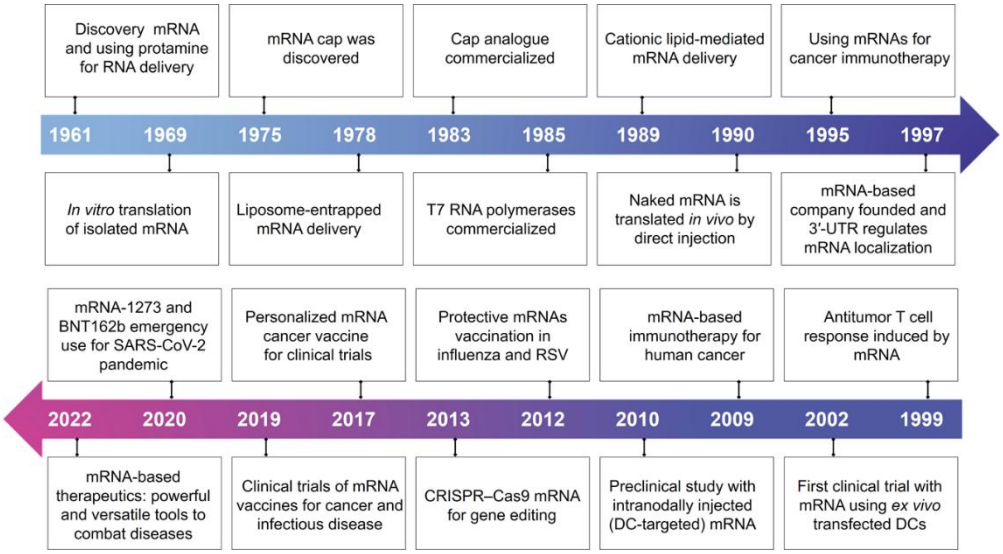


Figure 3. Significant findings and developments in mRNA-based treatments (Qin et al., 2022)

4.2 Production of Synthetic mRNA

Therapeutic mRNA is typically generated through *in vitro* transcription (IVT). A linear DNA template, which includes the 5′ cap, UTRs, coding sequence, and poly(A) tail, is transcribed using bacteriophage RNA polymerases such as T7, T3, or SP6. Following this, the mRNA undergoes extensive purification, often employing chromatographic methods like reverse-phase high-performance liquid chromatography (RP-HPLC), to remove double-stranded RNA and incomplete transcripts that could trigger undesired immune reactions (Baiersdörfer et al., 2019). Finally, the purified mRNA is packaged into lipid nanoparticles (LNPs), which protect it from enzymatic degradation and enable efficient delivery into target cells.

4.3 Advantages of mRNA Therapy

mRNA therapeutics are inherently safer than DNA-based gene therapy because they do not need to enter the nucleus and carry no risk of genome integration, eliminating the possibility of insertional mutagenesis (Conry et al., 1995; Deal et al., 2021; Pardi et al., 2018)Unlike enzyme or protein-based therapies, mRNA allows sustained protein production for days to weeks. This approach is particularly promising for treating monogenic diseases, where protein synthesis is defective or absent. Compared to traditional protein replacement or gene therapy, mRNA therapy is cost-effective, versatile, and capable of inducing proteins that are properly folded and modified, offering more safer and personalized treatments for a wide range of diseases (Wei et al., 2023).

Delivery systems, especially targeted lipid-based nanoparticles, ensure the mRNA reaches the target cells effectively and can well protect the molecule from degradation, avoid immune detection, and enhance uptake. Preclinical studies have demonstrated the potential of mRNA therapy in treating a variety of conditions from lung diseases and asthma to liver disorders and single-gene deficiencies like alpha-1 antitrypsin deficiency or hemophilia by enabling cells to produce therapeutic proteins at functional levels (Michel et al., 2016).

5. mRNA Design and Optimization

The therapeutic performance of mRNA is strongly influenced by its structural design, which governs translational efficiency, stability, and immunogenicity. Key structural elements need to be optimized to mimic endogenous mRNA and promote efficient protein expression (Cao et al., 2025).

5.1 mRNA Capping

Eukaryotic mRNAs possess a 5′ cap structure formed by the attachment of a 7-methylguanosine (m^7G) residue via a triphosphate linkage to the first nucleotide of the transcript (Muttach et al., 2017). This cap structure is essential for mRNA stability and translation initiation by shielding the 5′ end from exonuclease-mediated degradation (Mugridge et al., 2018). It serving as a recognition site for the translation initiation factor eIF4E, the cap enables efficient recruitment of the ribosomal machinery and initiation of protein synthesis (Graifer & Karpova, 2020).

5.2 Role of Untranslated Regions (UTRs)

The untranslated regions flanking the coding sequence play an important role in regulating mRNA stability and translation efficiency. Although UTRs do not encode amino acids, they

interact with RNA-binding proteins and influence ribosomal recruitment, mRNA decay, and intracellular localization (Jia et al., 2020). The 5′UTR governs translation initiation and ribosomal scanning, whereas the 3′UTR primarily affects mRNA stability and translational persistence (Jia et al., 2020; Rasekhian et al., 2021).

5.3 Optimising Open Reading Frames

An individual organism's genome contains open reading frames (ORF), which can be sequences that code for proteins. In genes, ORFs are found between the start and stop codes (Martinez et al., 2020). Increased mRNA half-life is typically conferred by shorter ORFs, and organised coding sections can further improve stability (Wayment-Steele et al., 2021). A crucial factor for designing ORF is GC content. While very low GC content can cause premature transcription termination, a moderate GC enrichment stabilises mRNA structure and encourages effective translation (Chan et al., 2009). On the other hand, a very high GC content may produce stable secondary structures that activate innate immune sensors and obstruct ribosome scanning (Nürnberger & Brunner, 2002).

5.4 Poly(A) Tail Optimization

The majority of eukaryotic mRNAs have a poly(A) tail at the 3′ end that contains between 100 and 250 adenosine residues (Nakamura et al., 2020). This tail works with the 5′ cap to support translation initiation, improves mRNA stability, and guards against exonuclease degradation. Notably, a poly(A) tail of approximately 75 nucleotides was found to support the highest translation efficiency in human cell-free systems, coinciding with the predominant poly(A) length observed in endogenous human mRNAs (Biziaev et al., 2024).

6. Signal Peptides

Signal peptides are short amino-terminal stretches of roughly 15–25 amino acids that direct newly synthesized proteins toward the secretory pathway (Halic et al., 2006). These sequences facilitate the recognition and delivery of nascent secretory and membrane-bound polypeptides to the endoplasmic reticulum (ER) through a signal recognition particle (SRP)-dependent mechanism (Owji et al., 2018). Signal peptidase, which is found on the luminal side of the ER membrane, removes the signal peptide once translocation into the ER is started, allowing the polypeptide chain to continue elongating until synthesis is finished (Lumangtad & Bell, 2020). Previous studies have also demonstrated that variations in signal sequence composition can

significantly affect the expression levels of secretory proteins produced in cell-free or in vitro expression systems (Wen et al., 2011).

Despite having variability in length, signal peptides share a conserved structural organization consisting of three functional regions (Von Heijne, 1990). The amino-terminal N-region is typically enriched in positively charged residues, which are important for initial interactions with the translocation machinery. This is followed by the hydrophobic H-region, which forms the core of the signal peptide and is essential for membrane insertion. The carboxy-terminal C-region contains the signal peptidase cleavage site that allows removal of the signal peptide once translocation into the ER has occurred (Von Heijne, 1990). Although this general architecture is conserved, substantial differences in amino acid composition are observed among signal peptides, contributing to their functional diversity.

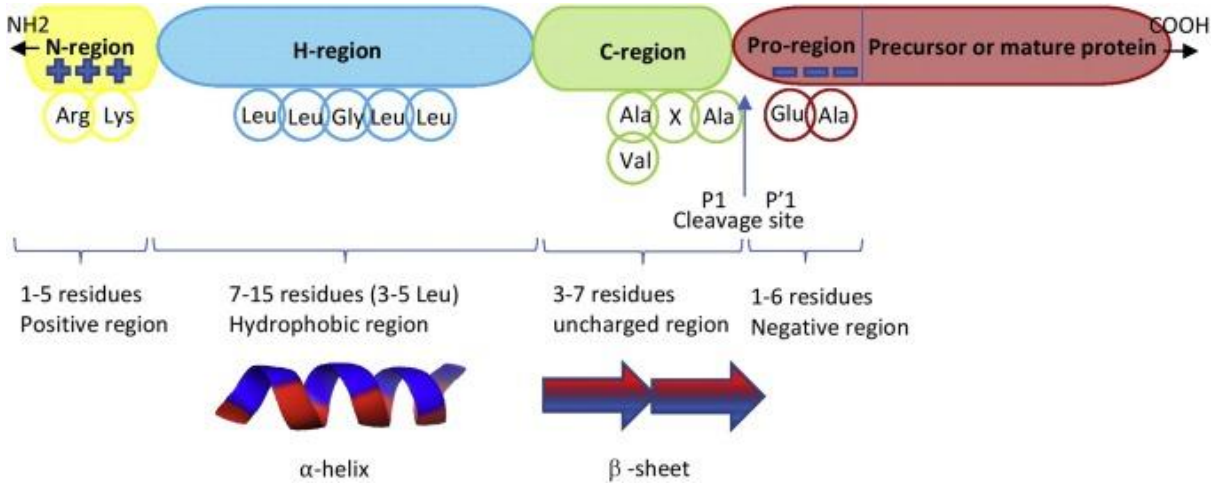


Figure 4. General structure of signal peptide (Owji et al., 2018)

6.1 Importance of Signal Peptide-Mediated Entry

Biological activity of α-Galactosidase A depends on accurate targeting to the secretory pathway during synthesis. The enzyme is produced as a precursor polypeptide that includes an N-terminal signal peptide, which is essential for directing the nascent protein to the endoplasmic reticulum (ER). During translation, this signal sequence enables ribosome association with the ER membrane, ensuring translocation of the growing polypeptide into the ER lumen where folding and maturation occur (Pool, 2022). Proteins that fail to engage the ER translocation machinery are prone to cytosolic misfolding and rapid degradation (Fasana et al., 2024). Therefore, the signal peptide plays a significant role that determines how much newly synthesized GLA becomes available for secretion and lysosomal delivery.

7 mRNA Therapy for Fabry Disease

Two independent mRNA-based therapeutic programs have been reported for Fabry disease, led by Translate Bio and Moderna, respectively. Both approaches were designed to evaluate whether systemic delivery of lipid nanoparticle (LNP) encapsulated mRNA encoding α-galactosidase A could enable endogenous enzyme production and restore enzymatic activity in relevant tissues.

Table 2. mRNA therapies and its current status- Fabry disease

Company	Models Used	Dose & Design	Key Findings	Main Limitation	Current Status Study	Study
Translate Bio	Mice & non-human primates	1 mg/kg IV α-Gal A mRNA-LNPs	Sustained enzyme secretion-activity in liver, heart, kidney	Secretion efficiency and use of signal peptide was not investigated	Preclinical stage	(DeRosa et al., 2019)
Moderna	Mice & non-human primates	0.5 mg/kg IV codon-optimized α-GalA mRNA-LNPs	Long-lasting enzyme expression	No direct comparison with ERT & determinants of secretion not investigated	Discontinued	(Zhu et al., 2019)

9. Research Gap

1. High treatment cost and lifelong dependence make enzyme replacement therapy inaccessible for many patients.
2. Limited enzyme stability and tissue delivery necessitate frequent dosing.
3. Therapeutic efficacy is reduced by immune reactions against administered enzymes.
4. Limited understanding of how signal peptide selection influences mRNA-encoded lysosomal enzyme secretion
5. Insufficient characterization of post-translational quality of secreted GLA following signal-peptide modification

10. Hypothesis

Altering the N-terminal signal peptide of α-Galactosidase A mRNA is expected to modulate enzyme secretion and processing in producer cells, thereby providing a framework to evaluate how secreted enzyme from conditioned media can restore intracellular GLA activity and support functional recovery in recipient cells.

12. Research Objectives

1. To construct and evaluate a panel of α-Galactosidase A (GLA) mRNA variants incorporating different N-terminal signal peptides.
2. To characterize the biochemical properties of the secreted enzyme obtained from each signal-peptide variant
3. To assess the ability of secreted enzyme to restore GLA activity and support functional recovery in recipient cell models

13. Proposed Workplan

1.Design and evaluation of GLA mRNA constructs

1.1 Design and synthesis of a set of GLA mRNA constructs encoding different N terminal signal peptides such as native GLA, tPA, and up to five additional variants.

1.2 Transfection of HEK293 cells with equal amounts of each GLA mRNA construct.

1.3 Collection of cell lysates and conditioned media at different timepoints post transfection.

1.4 Quantification of α Galactosidase A activity using a fluorometric 4 MU assay along with protein abundance analysis by western blotting.

2. Biochemical characterization of secreted α Galactosidase

2.1 Collection of conditioned media from the top performing signal peptide variants identified.

2.2 Analysis of post translational processing and glycosylation using endoglycosidase digestion followed by SDS PAGE and western blotting.

2.3 Evaluation of enzyme stability for extended durations and measurement of residual enzymatic activity at each time point.

3. Functional assessment of cross correction mediated by secreted α Galactosidase A

3.1 Application of conditioned media from top performing signal peptide variants to recipient cell lines.

3.2 Measurement of intracellular α Galactosidase A activity at pre-decided timepoints post treatment.

3.3 Confirmation of enzyme uptake and subcellular localization using immunofluorescence staining with anti GLA and LAMP1 antibodies.

Table 3. Signal Peptide sequences and its origin

S.no.	Amino acid Sequence	Signal Peptide Origin	Reference
1	MQLRNPELHLGCALALRFLALVSWDIPGARA	Native GLA signal peptide	(Tsuji et al., 1987)
2	MGTVRSRRLWWPLPLLLLLLRGPAGARA	N-terminal signal peptide of black-handed spider monkey proprotein convertase subtilisin/kexin type 9.	(O'Neill et al., 2023)
3	MDFQVQIISFLLISASVIMSRG	N-terminal signal peptide expressing rituximab LC in CHO K1. Published name: rituximab native LC.	(O'Neill et al., 2023)
4	MAFLWLLSCWALLGTTFG	Chymotrypsinogen B1	(Cheng et al., 2021)

14. Expected Outcomes

1. Identification of signal peptides that enhance extracellular secretion of mRNA-encoded α-Galactosidase A
2. Clear evidence that signal-peptide choice differentially affects intracellular expression and secretion efficiency
3. Effective characterization of post-translational quality of secreted GLA
4. Validation of functional cross-correction, with secreted GLA restoring intracellular activity in recipient cells

15. Conclusion

mRNA based therapeutic strategies provide an effective platform for intracellular production of therapeutic proteins, but their success depends on efficient targeting, processing, and secretion of the expressed enzyme. The role of signal peptide selection in shaping these downstream events remains insufficiently characterized, particularly for mRNA encoded lysosomal enzymes such as α-Galactosidase A.

This study will systematically examine how variation in N terminal signal peptides influences the expression, secretion, and biochemical quality of α-Galactosidase A. By combining quantitative secretion analysis with detailed evaluation of post translational processing and enzyme stability, the work aims to identify signal peptide designs that enhance enzyme availability while preserving functional integrity. Functional cross correction assays further assess the ability of secreted enzyme to restore intracellular activity in recipient cell models.

Given the multisystem nature of Fabry disease, improving extracellular delivery of functional enzyme remains a critical challenge. The findings from this study are expected to provide practical insights into signal peptide mediated optimization of mRNA therapeutics and contribute to the improved treatment strategies. This work may also have broader relevance for other lysosomal storage disorders that rely on secretion driven cross correction for therapeutic benefit.

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